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Final Report

VRT-GRPO: Explicit Virtual-Region Reasoning for Multi-Entity Visual Grounding in Remote-Sensing Images

by

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repository: https://github.com/ese-ada-lovelace-2025/irp-sy1124.git

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Abstract

With the development of human-computer interaction and embodied intelligence, accurately mapping complex natural language constraints to targets in physical space has become a key capability for Vision-Language Models (VLMs). Remote sensing images, characterized by wide coverage, dense targets, and complex spatial relationships, provide an ideal scenario for studying this capability. However, existing VLM-based remote sensing visual grounding methods primarily model targets and reference objects. Several studies show that explicit intermediate reasoning states can mitigate the "attention drift" phenomenon in long-sequence decoding. Inspired by this, this paper proposes VRT-GRPO: utilizing explicit virtual regions as intermediate states to simulate human cognitive paths, transforming abstract spatial relationships into verifiable and optimizable region representations, and optimizing each stage via GRPO with process rewards. The mAcc@0.5 metric is adopted across multiple VLMs to evaluate localization performance at each stage. The performance differences across various backbone models indicate that VRT-GRPO significantly improves target localization performance on VLMs with stronger instruction-following capabilities, demonstrating sensitivity to the backbone model's instruction-following ability.

Keywords: Vision-Language Models; Remote Sensing Visual Grounding; Spatial Reasoning; Explicit Virtual Region; Process Reward; GRPO.

1 Introduction

With the continued development of vision-language models (VLMs) in human–computer interaction, intelligent remote-sensing interpretation, and embodied intelligence, accurately mapping complex natural-language constraints to physical targets in images has become an important problem in multimodal understanding [1–3]. Visual grounding aims to localize the image region corresponding to a natural-language expression, allowing users to describe targets of interest through categories, attributes, locations, and spatial relations [4–9]. This capability is useful not only for remote-sensing image retrieval and scene understanding [8–10], but also as a foundation for applications such as embodied navigation and human–robot interaction [2, 3, 11].

Remote-sensing images provide a representative setting for studying complex spatial-language understanding. Compared with natural images, they typically cover larger areas, exhibit greater scale variation, and contain many visually similar and densely distributed repeated targets [8–10]. A model often cannot identify a unique instance from category or appearance alone and must use spatial relations for disambiguation [4, 9, 12]. For example, given "the basketball court to the right of the lake," the model must not only recognize the lake and basketball courts, but also understand how "to the right of" constrains the court's possible location and exclude candidates that do not satisfy the relation. Therefore, complex visual grounding in remote sensing requires not only recognizing the target and reference entity, but also understanding how spatial relations constrain the target search space [4, 12].

Research on remote-sensing visual grounding has progressed from holistic cross-modal matching between language expressions and image regions toward finer-grained use of target attributes, spatial relations, and scene context [8–10, 13, 14]. More recently, Think and Answer ME constructed the ME-RSRG dataset and proposed the Entity-Aware Reasoning (EAR) framework, extending remote-sensing visual grounding from single-entity matching to joint multi-entity grounding [12]. EAR explicitly generates bounding boxes for the subject (target) and reference object, together with corresponding textual reasoning, enabling joint modeling of the target and its reference and providing an important foundation for complex relation-aware grounding

[12].

However, explicit entity modeling does not amount to explicit spatial modeling of relations. Given a reference entity and a spatial relation, we argue that the direct spatial consequence of the relation is to constrain the candidate region in which the target may appear. Accordingly, a key aspect of spatial-relation understanding is whether the relation can be projected into a localizable and verifiable spatial constraint. Existing methods are still generally optimized around the final target region or entity boxes, while spatial relations mainly contribute as language features, textual reasoning, or reward conditions and are rarely represented as independently generatable, evaluable, and optimizable spatial intermediate states [4, 12–14].

Based on this observation, we propose VRT-GRPO, a stage-wise spatial reasoning framework with an explicit virtual region as an intermediate state. VRT-GRPO first generates a semantic plan and then sequentially performs reference-entity localization, virtual-region reasoning, and final target localization. The virtual region is a candidate search range jointly induced by the reference-entity location and the spatial relation. The model must therefore answer not only "where are the target and reference object?" but also "within which region should the target be searched for according to the spatial relation?" The overall comparison is illustrated in Figure 1.

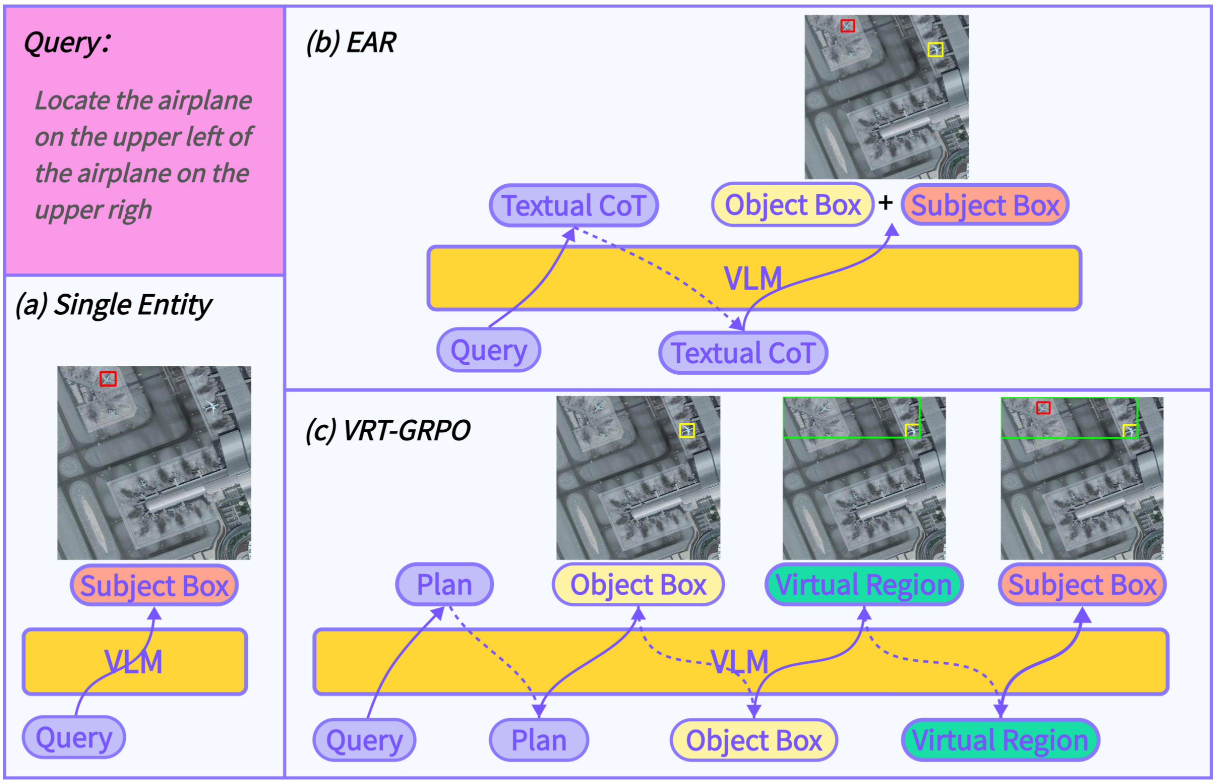

Figure 1: Comparison of direct visual grounding, textual chain-of-thought grounding, and the proposed VRT-GRPO spatial reasoning process grounding.

To supervise this intermediate state and evaluate the complete region-aware framework that incorporates it, we define rules for constructing virtual-region pseudo-annotations and use them to generate training data with region supervision. During training, SFT warm-up first teaches the model to follow the structured generation protocol [15]. Group Relative Policy Optimization (GRPO) is then applied with fine-grained rewards for different stages to optimize the complete reasoning trajectory, allowing relation-induced intermediate regions to be explicitly generated, evaluated, and optimized [16, 17].

We evaluate VRT-GRPO across different VLM backbones. The results show that, on the Qwen-Instruct family, the method improves reference-entity localization and virtual-region generation quality while yielding overall gains in grounding performance. Performance differences across backbones further indicate that the benefits of explicit spatial intermediate states depend on a model's ability to follow complex structured instructions, maintain stable long-sequence generation, and perform spatial reasoning, resulting in a degree of backbone sensitivity.

The main contributions of this work are as follows:

1. We formulate relation-aware visual grounding in remote sensing as a process in which spatial relations progressively constrain the search space, and introduce virtual regions together with rules for constructing their pseudo-labels.
2. We propose VRT-GRPO, a stage-wise spatial reasoning framework that uses structured generation trajectories and stage-specific process rewards to optimize the sequential process of reference-entity localization, virtual-region reasoning, and target localization.
3. We systematically evaluate VRT-GRPO across multiple VLM backbones, analyzing virtual-region generation quality, its role in entity localization, and backbone sensitivity, while revealing the discrepancy between spatial intermediate-state quality and final target-localization gains.

2 Related Work

2.1 Remote-Sensing Visual Grounding

Early research on remote-sensing visual grounding focused primarily on task definition, dataset construction, and cross-modal localization models. *Visual Grounding in Remote Sensing Images* initiated a systematic investigation of natural-language-guided target localization in remote-sensing imagery [8]. RSVG further constructed a large-scale benchmark for remote-sensing visual grounding and used hierarchical cross-modal feature learning to integrate multiscale visual features with multi-granularity textual representations, mitigating scale variation and background interference [9]. VRSBench incorporated referring expression comprehension, image captioning, and visual question answering into a large-scale remote-sensing vision-language benchmark, extending the training and evaluation of general-purpose VLMs in remote-sensing scenarios [10].

As grounding expressions increasingly incorporated complex attributes and spatial descriptions, research shifted from holistic cross-modal matching toward finer-grained use of linguistic cues. ProVG decouples language expressions into global context, spatial relations, and target attributes, and progressively modulates visual attention through a "survey–locate–verify" mechanism to achieve coarse-to-fine cross-modal alignment [13]. RSGround-R1 further introduces CoT supervision, position-aware rewards, and spatial-consistency optimization, using SFT and reinforcement fine-tuning to enhance position awareness and spatial reasoning [14]. Although these methods improve the use of spatial cues, their training and evaluation generally remain centered on the final target region, with spatial relations serving as feature modulation, textual reasoning, or reward conditions rather than explicit intermediate states [13, 14].

2.2 Relation-Aware and Multi-Entity Grounding

Complex visual-grounding expressions typically contain a target, reference entities, and attribute or spatial relations between them [4]. Although conventional methods use these relations to disambiguate the target, most output only the final target region, leaving reference entities and the relation-reasoning process implicit in cross-modal features [4–9]. To address this limitation, Think and Answer ME constructed the ME-RSRG dataset and proposed the Entity-Aware Reasoning (EAR) framework, reformulating remote-sensing visual grounding as a multi-entity reasoning-grounding task [12].

A key advance of EAR is its explicit distinction between entity roles and its separate evaluation of subject and reference-object localization. Its relational-consistency reward primarily checks whether the subject and at least one reference object are correctly matched simultaneously, with additional rewards for correctly matching multiple reference objects [12]. This design provides verifiable feedback for joint multi-entity grounding, but the reward does not directly assess relation direction or generate and evaluate the candidate region defined by the relation. In contrast, we introduce a virtual region jointly induced by the reference entity and spatial relation beyond the subject–object structure, making the relation-constrained search space a coordinate-based spatial intermediate state that can be independently evaluated and optimized.

2.3 Explicit Intermediate States and Process Supervision

Compared with supervising only final outcomes, process supervision provides finer-grained feedback for intermediate reasoning steps [18]. Chain-of-Thought decomposes complex problems by generating sequential textual steps [19], while Tree of Thoughts further allows models to explore, evaluate, and select among multiple candidate reasoning paths [20]. However, the intermediate states in these methods are carried primarily by natural language and cannot directly represent locations and regions in visual space [19, 20].

Multimodal reasoning research has therefore begun introducing visualized or coordinatized intermediate states. Visual Sketchpad enables models to use external visual sketches, such as lines, boxes, and markers, to support planning and reasoning [21]. MVoT introduces visual imagination into spatial reasoning by generating image visualizations corresponding to reasoning trajectories [22]. Grounded Chain-of-Thought further requires models to progressively localize relevant visual cues and use coordinates as visual evidence for answer reasoning [23]. Point-RFT binds reasoning steps to corresponding visual elements and optimizes visually grounded CoT through reinforcement fine-tuning [24]. These studies show that intermediate steps in multimodal reasoning can extend beyond pure text to explicit states linked to visual evidence.

2.4 Reinforcement Learning with Verifiable Rewards

Regarding optimization, GRPO estimates advantages from relative rewards within a group and optimizes the generation policy without training an additional value model [16]. Visual-RFT extends reinforcement fine-tuning with verifiable rewards to visual tasks and uses IoU-based visual rewards for tasks such as detection [17]. We follow the general idea of verifiable process optimization but focus on a candidate search region jointly determined by the reference-entity location and spatial relation. This design allows relation-induced spatial constraints to be explicitly generated and independently evaluated, while providing a basis for analyzing the relationship between intermediate-state quality and final target localization.

3 Methods

We propose VRT-GRPO, which formulates complex relation-aware remote-sensing visual grounding as a staged generation process that progressively narrows the search space through linguistic constraints. Its core idea is to explicitly represent the candidate search range jointly constrained by the reference entities and spatial relations as a virtual region and place it between reference-object localization and final target localization. In this way, spatial relations are no longer conveyed solely through natural-language reasoning but instead form a coordinate-based intermediate state that can be generated, evaluated, and optimized.

The overall method comprises three stages: virtual-region pseudo-annotation construction, structured-trajectory SFT initialization, and region-aware GRPO optimization.

3.1 Task Definition

Given a remote-sensing image I and a relation-aware natural-language query q, the model is required to localize the target entity specified by the query [5, 6, 8, 9]. Following the entity partition used in multi-entity relational grounding, we refer to the final target to be localized as the subject and the entities used to describe its location as reference objects [4, 12]. For example, in "the basketball court to the right of the lake," *basketball court* is the subject, *lake* is the reference object, and *to the right of* denotes the spatial relation between them.

Let the ground-truth bounding box of the subject be b_s^*, and let the set of ground-truth bounding boxes of the reference objects be

$$\mathcal{B}_o^* = \{b_o^{1*}, b_o^{2*}, \dots, b_o^{N*}\}. \tag{1}$$

Existing multi-entity grounding methods typically predict the subject box b_s and the set of reference-object boxes $\mathcal{B}_o$ [12]. We further introduce a virtual region b_r, extending the model output to

$$y = (p, \mathcal{B}_o, b_r, b_s), \tag{2}$$

where p denotes the semantic plan, $\mathcal{B}_o$ denotes the predicted set of reference-object boxes, b_r denotes the predicted virtual region, and b_s denotes the final subject box.

3.2 Overall VRT-GRPO Framework

VRT-GRPO explicitly models the continuous process of "reference-object localization $\rightarrow$ spatial-region reasoning $\rightarrow$ target localization" during generation.

Specifically, the model first generates a semantic plan from the query to identify the target, reference entities, and spatial relations. It then localizes the reference objects, generates a virtual region based on the reference entities and relational constraints, and finally predicts the subject coordinates under the constraint of this candidate region. This process can be formulated as

$$(I, q) \rightarrow p \rightarrow \mathcal{B}_o \rightarrow b_r \rightarrow b_s. \tag{3}$$

Owing to the autoregressive nature of large language models, final subject localization is conditioned at the sequence level on the semantic plan, reference-object coordinates, and virtual-region coordinates [25]. The complete VRT-GRPO training workflow is illustrated in Figure 2.

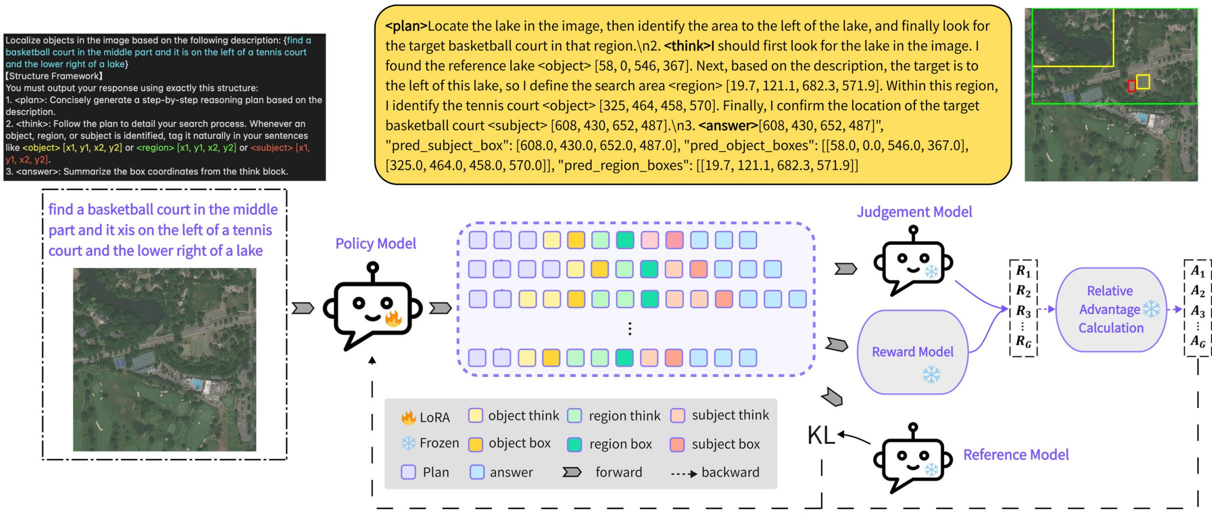

Figure 2: Overview of the VRT-GRPO training workflow. The policy model generates structured trajectories containing planning, reference-object, virtual-region, subject, and answer states. A frozen judgment model and reward model evaluate the sampled trajectories to produce group-relative advantages, while a frozen reference model provides KL regularization during policy optimization.

3.3 Virtual-Region Pseudo-Annotation Construction

To enable spatial relations to participate in training as explicit regions, we construct virtual-region pseudo-annotations from the existing subject and reference-object annotations.

Given a subject bounding box b_s and a set of reference-object bounding boxes $\mathcal{B}_o = \{b_o^i\}_{i=1}^N$, we first calculate the aggregate center of all reference objects and the center of the subject, denoted by $c_o = (c_o^x, c_o^y)$ and $c_s = (c_s^x, c_s^y)$, respectively. We also define

$$b_u = \mathrm{BBox}(b_s \cup b_o^1 \cup \cdots \cup b_o^N) \tag{4}$$

as the joint enclosing box covering the subject and all reference objects, where

$$b_u = [x_1^u, y_1^u, x_2^u, y_2^u]. \tag{5}$$

We then construct a directional candidate region from the joint enclosing box according to the direction of c_s relative to c_o. The central idea is to use the reference-object center as the boundary or corner on the side closer to the anchor and extend the region toward the subject. Specifically, the horizontal and vertical ranges of the initial virtual region $\tilde{b}_r = [\tilde{x}_1, \tilde{y}_1, \tilde{x}_2, \tilde{y}_2]$ are defined as

$$[\tilde{x}_1, \tilde{x}_2] = \begin{cases} [c_o^x, x_2^u], & c_s^x > c_o^x, \\ [x_1^u, c_o^x], & c_s^x \le c_o^x, \end{cases} \tag{6}$$

while the vertical range is determined analogously from the relative positions of c_s^y and c_o^y. Finally, a small amount of padding is added to $\tilde{b}_r$, and the result is clipped to the image boundaries to obtain the final virtual-region pseudo-annotation:

$$b_r = \mathrm{Clip}(\mathrm{Pad}(\tilde{b}_r, \lambda), I). \tag{7}$$

This construction can be viewed as a center-guided directional region-generation strategy. Compared with directly using the joint enclosing box that merely covers all subjects and objects, the virtual region better emphasizes the search direction from the reference object toward the subject. This pseudo-region serves as the basis for subsequent trajectory supervision and region-level rewards. The construction process is illustrated in Figure 3.

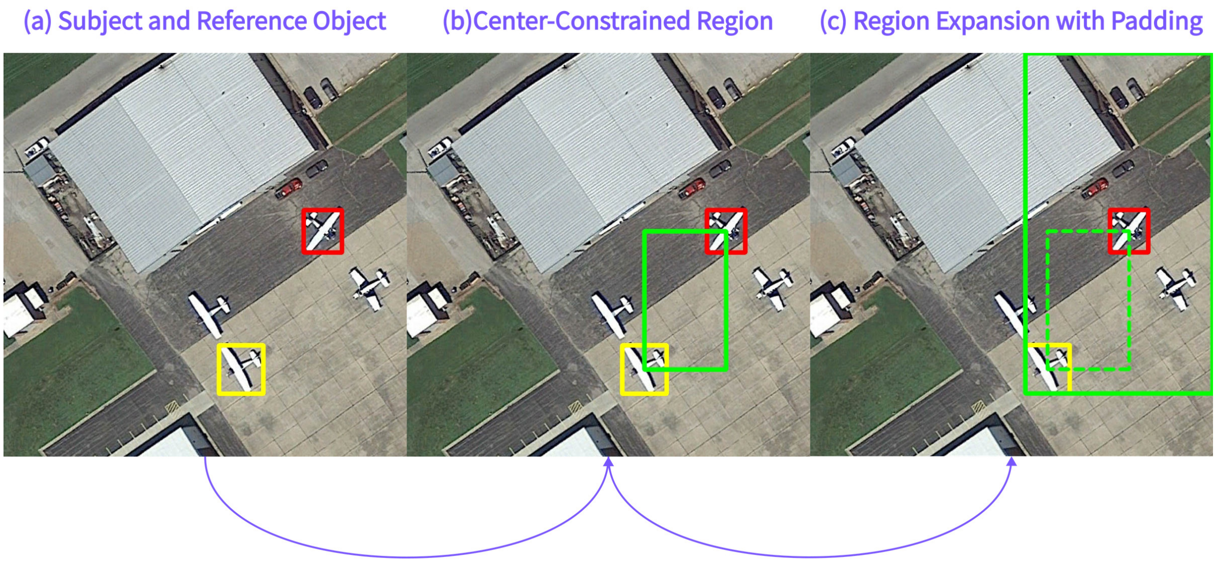

Figure 3: Illustration of the virtual-region construction process. The green dashed box denotes the center-constrained region formed by the center points of the subject and reference object, and the green solid box denotes the final virtual region obtained after directional expansion and padding.

Algorithm 1 Virtual Region Construction

Require: Subject box b_s, reference-object boxes $\mathcal{B}_o = \{b_o^i\}_{i=1}^N$, image size (W, H), padding ratio λ

Ensure: Virtual region pseudo-label b_r

Compute the union box $b_u \leftarrow \mathrm{BBox}(\{b_s\} \cup \mathcal{B}_o)$
Compute the reference union box $b_o^U \leftarrow \mathrm{BBox}(\mathcal{B}_o)$
Compute centers $c_s \leftarrow \mathrm{Center}(b_s)$ and $c_o \leftarrow \mathrm{Center}(b_o^U)$
Let $b_u = [x_1^u, y_1^u, x_2^u, y_2^u]$
for $d \in \{x, y\}$ **do**
  Let $[l_d^u, u_d^u]$ be the interval of b_u along dimension d
  if $c_s^d > c_o^d$ **then**
    Set the directional interval $\tilde{I}_d \leftarrow [c_o^d, u_d^u]$
  else
    Set the directional interval $\tilde{I}_d \leftarrow [l_d^u, c_o^d]$
  end if
end for
Form the directional region $\tilde{b}_r \leftarrow \tilde{I}_x \times \tilde{I}_y$
$b_r \leftarrow \mathrm{Clip}(\mathrm{Pad}(\tilde{b}_r, \lambda), W, H)$
return b_r

3.4 Trajectory Supervision and SFT Warm-up

After obtaining the virtual-region pseudo-annotations, we organize each training sample as a structured generation trajectory containing intermediate spatial states. Given an image I and query q, the target output sequence is represented as

$$Y = (Y_{\text{plan}}, Y_{\text{obj}}, Y_{\text{reg}}, Y_{\text{sub}}, Y_{\text{ans}}), \tag{8}$$

where Y_{plan} denotes semantic planning; Y_{obj}, Y_{reg}, and Y_{sub} denote the coordinate generation of the reference objects, virtual region, and subject, respectively; and Y_{ans} summarizes the final answer.

The textual format is as follows:

```
<plan> ... </plan>
<think>
... <object_1> [x1, y1, x2, y2] ...
...
... <object_n> [x1, y1, x2, y2] ...
...
... <region> [x1, y1, x2, y2] ...
...
... <subject> [x1, y1, x2, y2] ...
</think>
<answer> object: [...], region: [...], subject: [...] </answer>
```

Here, `<plan>` describes the target, reference entities, and spatial relations; `<think>` generates the `<object>`, `<region>`, and `<subject>` coordinates in the reasoning order; and `<answer>` summarizes the coordinate results.

We use SFT to initialize the learning of this structured trajectory, enabling the model to establish stable role distinctions and generation order at the beginning of training. This stage primarily alleviates the cold-start problem of direct GRPO optimization and provides a stable initial policy for subsequent process-reward optimization [15–17].

3.5 Region-Aware GRPO Optimization

After SFT initialization, we apply GRPO to optimize the core reasoning trajectory. For the same input (I, q), the model samples a group of candidate outputs $\{y_k\}_{k=1}^{K}$, and each trajectory is evaluated using the reward function $R(y_k)$. GRPO estimates advantages from within-group relative rewards, thereby optimizing the generation policy without training an additional critic model [15, 16].

The overall reward consists of five types of signals:

$$\begin{aligned} R = {} & \lambda_{\text{plan}} R_{\text{plan}} + \lambda_{\text{fmt}} R_{\text{fmt}} + \lambda_{\text{ent}} R_{\text{ent}} \\ & + \lambda_{\text{reg}} R_{\text{reg}} + \lambda_{\text{spa}} R_{\text{spa}}. \end{aligned} \tag{9}$$

The format reward constrains the output format:

$$R_{\text{fmt}} = R_{\text{str}} + R_{\text{num}} + R_{\text{cons}}. \tag{10}$$

Here, R_{str} checks whether the output contains complete and correctly ordered `<plan>`, `<think>`, and `<answer>` tags; R_{num} requires `<think>` to contain exactly one subject, exactly one region, and the same number of reference objects as in the annotations; and R_{cons} checks whether the coordinates in `<think>` and `<answer>` are consistent. This reward ensures a stable format for subsequent geometric evaluation.

The entity-localization reward evaluates entity-localization quality, including subject localization and reference-object grounding:

$$R_{\text{ent}} = R_{\text{sub}} + R_{\text{obj}}. \tag{11}$$

For the subject, the reward is calculated from the IoU between the predicted and ground-truth boxes [17, 26]:

$$R_{\mathrm{sub}} = f_s(\mathrm{IoU}(\hat{b}_s, b_s^*)). \tag{12}$$

For reference objects, the predicted and ground-truth boxes are aligned through one-to-one matching, denoted by $\pi(i)$, and the average reward is calculated as [27]

$$R_{\mathrm{obj}} = \frac{1}{N}\sum_{i=1}^{N} f_o(\mathrm{IoU}(\hat{b}_o^{\pi(i)}, b_o^{i*})). \tag{13}$$

Here, f_s and f_o are piecewise IoU-based reward functions that encourage predictions with high overlap and penalize clearly incorrect localization results.

The virtual-region reward evaluates whether the predicted region forms a valid candidate search space. Let the predicted region be $\hat{b}_r$ and the pseudo-annotated region be b_r^*. Then,

$$R_{\mathrm{reg}} = 1.5\,\mathrm{IoU}(\hat{b}_r, b_r^*) + 0.5\,\mathrm{Cov}(b_s^*, \hat{b}_r) + \Delta_{\mathrm{reg}}, \tag{14}$$

where

$$\mathrm{Cov}(b_s^*, \hat{b}_r) = \frac{|b_s^* \cap \hat{b}_r|}{|b_s^*|}. \tag{15}$$

This reward simultaneously encourages the region to approximate the pseudo-annotated region and cover the final subject, allowing it to function as a candidate search range.

The spatial-consistency reward further checks whether the predicted region lies in a plausible relational direction. Let $\mathcal{D}(q)$ denote the directional dimensions involved in the query, such as horizontal or vertical. For the i-th reference object, directional consistency is defined as

$$\begin{aligned} a_i = \frac{1}{|\mathcal{D}(q)|}\sum_{d\in\mathcal{D}(q)} &\mathbb{I}\big[\mathrm{sign}(c_d(\hat{b}_r) - c_d(b_o^{i*})) \\ &= \mathrm{sign}(c_d(b_s^*) - c_d(b_o^{i*}))\big], \end{aligned} \tag{16}$$

where $c_d(\cdot)$ denotes the center coordinate of a bounding box along dimension d. The spatial-consistency reward is

$$R_{\mathrm{spa}} = \mathrm{Cov}(b_s^*, \hat{b}_r) + \Delta_{\mathrm{cov}} + 0.5\left(2 \cdot \frac{1}{N}\sum_{i=1}^{N} a_i - 1\right). \tag{17}$$

This term does not require the predicted region to completely overlap the pseudo-annotation; instead, it emphasizes whether the region covers the target and follows the direction specified by the spatial relation.

The planning reward R_{plan} supplements semantic constraints that are difficult to capture with geometric rewards [18]. We use a frozen external judge model to evaluate, at the textual level, how well <plan> covers the target, reference entities, attributes, and spatial relations in the query, as well as whether the subsequent <think> and <answer> follow the plan. Let the judge model provide a constraint-coverage score s_c and an execution-consistency score s_e, each ranging from $[0, 9]$. Then,

$$R_{\mathrm{plan}} = \frac{s_c + s_e - 12}{6}. \tag{18}$$

Through this reward design, VRT-GRPO distributes reinforcement-learning signals across different stages of the structured reasoning trajectory. The model therefore optimizes not only final subject localization but also reference-object localization, candidate-region generation, spatial-direction consistency, and adherence to the semantic plan.

Algorithm 2 VRT-GRPO Training Pipeline

Require: Base VLM π_θ, training set $D = \{(I_i, q_i, b^*_{s,i}, \mathcal{B}^*_{o,i})\}_{i=1}^{M}$, group size G
Ensure: Optimized region-aware grounding model π^*_θ

Initialize $D_{\mathrm{SFT}} \leftarrow \varnothing$ and $D_{\mathrm{RL}} \leftarrow \varnothing$
for $i = 1$ to M **do**
  $(W_i, H_i) \leftarrow \mathrm{Size}(I_i)$
  $b^*_{r,i} \leftarrow \mathrm{VirtualRegionConstruction}(b^*_{s,i}, \mathcal{B}^*_{o,i}, W_i, H_i, \lambda)$
  Construct structured trajectory $Y_i = (Y_{\mathrm{plan}}, Y_{\mathrm{obj}}, Y_{\mathrm{reg}}, Y_{\mathrm{sub}}, Y_{\mathrm{ans}})$ from $(q_i, \mathcal{B}^*_{o,i}, b^*_{r,i}, b^*_{s,i})$
  $D_{\mathrm{SFT}} \leftarrow D_{\mathrm{SFT}} \cup \{(I_i, q_i, Y_i)\}$
  $D_{\mathrm{RL}} \leftarrow D_{\mathrm{RL}} \cup \{(I_i, q_i, b^*_{s,i}, \mathcal{B}^*_{o,i}, b^*_{r,i})\}$
end for
$\pi_{\mathrm{SFT}} \leftarrow \mathrm{SFT}(\pi_\theta, D_{\mathrm{SFT}})$
Initialize policy $\pi_\theta \leftarrow \pi_{\mathrm{SFT}}$ and reference policy $\pi_{\mathrm{ref}} \leftarrow \pi_{\mathrm{SFT}}$
while not converged **do**
  Sample a batch $\mathcal{B} \sim D_{\mathrm{RL}}$
  for all $(I, q, b^*_s, \mathcal{B}^*_o, b^*_r) \in \mathcal{B}$ **do**
    Sample G trajectories $\{y_j\}_{j=1}^{G} \sim \pi_\theta(\cdot \mid I, q)$
    Parse each y_j into plan, objects, region, and subject
    Compute trajectory rewards R_j using $R_{\mathrm{plan}}, R_{\mathrm{fmt}}, R_{\mathrm{ent}}, R_{\mathrm{reg}}, R_{\mathrm{spa}}$
    Compute group-relative advantages $A_j = (R_j - \mu_R)/(\sigma_R + \epsilon)$
  end for
  Update π_θ with the GRPO objective
end while
return $\pi^*_\theta \leftarrow \pi_\theta$

4 Experiments

4.1 Dataset

We conduct experiments on the ME-RSRG dataset [12]. ME-RSRG is a benchmark for multi-entity reasoning grounding in remote sensing images. It contains 7,162 remote sensing images and 12,091 image-text instances, officially divided into 10,305 training instances and 1,786 test instances. Following the official setting, the train and validation splits are combined for training, and results are reported on the test split.

We follow the official training and test splits of ME-RSRG. In our implementation, the SFT stage uses 2,149 training samples with structured trajectories, the GRPO stage uses 8,156 training samples, and the test set contains 1,786 samples, all following the official ME-RSRG split.

4.2 Experimental Settings

We compare two reasoning paradigms. The first is the entity-aware baseline, in which the model generates textual reasoning and predicts the bounding boxes of the subject and reference objects without explicitly generating a virtual region [12]. The second is our proposed VRT paradigm, which introduces semantic planning and a virtual region into entity-aware grounding to form a structured plan–object–region–subject reasoning trajectory.

We use Qwen3-VL-8B-Instruct, Qwen2.5-VL-7B-Instruct, and InternVL-8B as backbone models [28–30]. Each model is evaluated at the Prompt, SFT, and GRPO stages. At the Prompt stage, the corresponding prompt is used directly for zero-shot inference without parameter updates. At the SFT stage, the model is supervised fine-tuned using structured trajectories. At the GRPO stage, optimization continues from the SFT LoRA adapter. The baseline and VRT use the same test set, decoding strategy, and evaluation metrics.

SFT is performed with LoRA on the `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` modules [31]. The GRPO stage is initialized from the SFT LoRA adapter and trained for one epoch on four H20 GPUs. Each optimizer update processes four prompts, with $K = 8$ rollouts sampled per prompt, producing 32 completions in total. The method uses the DAPO loss [32]. The reward function consists of five components: plan rationality, format and logical consistency, entity localization, region quality, and spatial-direction consistency, with weights of 0.15, 0.25, 1.00, 0.25, and 0.20, respectively. The external judge model is `qwen3vl-30b-critic`. During testing, greedy decoding is used uniformly with `do_sample=False` and a maximum generation length of 2,048 tokens.

4.3 Evaluation Metrics

The main task is evaluated using `Acc@0.5_sub`, `Acc@0.5_obj`, and `meanAcc@0.5`. For a dataset containing N test samples, the subject localization of sample i is considered correct if the IoU between the predicted subject box $\hat{B}_{s,i}$ and the ground-truth subject box $B_{s,i}$ is at least 0.5 [26]:

$$\mathrm{Acc@0.5_{sub}} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\Big[\mathrm{IoU}(\hat{B}_{s,i}, B_{s,i}) \geq 0.5\Big]. \tag{19}$$

For reference objects, predicted and ground-truth boxes are matched one-to-one [27]. Let m_i denote the number of correct matches with an IoU of at least 0.5 in sample i, and let n_i denote the number of ground-truth reference objects. Then,

$$\mathrm{Acc@0.5_{obj}} = \frac{\sum_{i=1}^{N} m_i}{\sum_{i=1}^{N} n_i}. \tag{20}$$

The combined metric is defined as

$$\mathrm{meanAcc@0.5} = \frac{\mathrm{Acc@0.5_{sub}} + \mathrm{Acc@0.5_{obj}}}{2}. \tag{21}$$

To analyze whether the model can generate an effective virtual region, we also evaluate region-related metrics. Let $\mathcal{V}$ denote the set of samples with valid regions. `Subject-cover@`τ is computed over this subset and measures whether the predicted region covers the ground-truth subject by at least a threshold τ. Coverage is calculated using the intersection over subject area:

$$\mathrm{IoC}(B_s, B_r) = \frac{|B_s \cap B_r|}{|B_s|}, \tag{22}$$

where B_s is the ground-truth subject box and B_r is the predicted region box. Therefore,

$$\text{Subject-cover@}\tau = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}}\mathbb{I}[\mathrm{IoC}(B_{s,i}, B_{r,i}) \geq \tau]. \tag{23}$$

`Cover+direction` is likewise computed over the subset of valid regions. A sample is considered correct only when the predicted region satisfies `Subject-cover@0.5` and its direction relative to the reference objects is consistent with that of the ground-truth subject.

4.4 Main Results and Stage-wise Analysis

Table 1 presents the main experimental results and compares performance across the Prompt, SFT, and GRPO stages. This analysis examines the performance of the VRT paradigm at different optimization stages.

Table 1: Grounding performance across different VLM backbones and training stages.

Backbone	Metric	Prompt		SFT		GRPO	
		Baseline	Ours	Baseline	Ours	Baseline	Ours
Qwen3-VL-8B-Instruct	Acc@0.5_sub	0.2682	0.0213	0.3376	0.3251	0.3796	0.4067
	Acc@0.5_obj	0.1695	0.0095	0.3489	0.3504	0.3694	0.4242
	meanAcc@0.5	0.2172	0.0154	0.3434	0.3378	0.3743	0.4155
Qwen2.5-VL-7B-Instruct	Acc@0.5_sub	0.1131	0.0146	0.3175	0.3108	0.4440	0.4927
	Acc@0.5_obj	0.0944	0.0073	0.3484	0.4744	0.4890	0.5172
	meanAcc@0.5	0.1035	0.0109	0.3334	0.3926	0.4672	0.5050
InternVL-8B	Acc@0.5_sub	0.0235	0.0140	0.3533	0.2243	0.3908	0.2974
	Acc@0.5_obj	0.0231	0.0047	0.3232	0.2611	0.3489	0.3295
	meanAcc@0.5	0.0233	0.0094	0.3378	0.2427	0.3692	0.3135

At the Prompt stage, VRT performs substantially worse than the baseline on all three models. For example, `meanAcc@0.5` decreases from 0.2172 to 0.0154 on Qwen3-VL-8B-Instruct and from 0.1035 to 0.0109 on Qwen2.5-VL-7B-Instruct. This indicates that, without training, the models struggle to follow the complex structured output protocol required by VRT. Therefore, the Prompt-stage results mainly reflect the zero-shot difficulty of following different output protocols.

The SFT-stage results show clear differences across models. On Qwen3-VL, overall performance remains close to that of the baseline, with `meanAcc@0.5` decreasing slightly from 0.3434 to 0.3378. Qwen2.5-VL mainly improves in reference-object localization, with `Acc@0.5_obj` increasing from 0.3484 to 0.4744. In contrast, all three metrics decrease on InternVL. These results indicate that different backbones vary in their ability to adapt to structured-trajectory supervision.

After reinforcement-learning optimization, VRT outperforms the corresponding baseline on both Qwen backbones. On Qwen3-VL-8B-Instruct, VRT improves `meanAcc@0.5` by 4.12%. On Qwen2.5-VL-7B-Instruct, VRT improves `meanAcc@0.5` by 3.78%. This shows that, on Qwen-Instruct models specifically tuned for instruction following, the complete VRT-GRPO framework with virtual regions and stage-specific process rewards improves multi-entity localization performance.

In contrast, InternVL-8B still gains no advantage at the GRPO stage. Relative to the baseline, VRT decreases `Acc@0.5_sub` by 9.34 percentage points, `Acc@0.5_obj` by 1.94 percentage points, and `meanAcc@0.5` by 5.57 percentage points. This result shows that the benefits of VRT do not generalize uniformly across backbone models. Whether structured intermediate states translate into final localization gains still depends on the model's instruction-following, reasoning, and grounding capabilities.

4.5 Virtual-Region Quality Analysis

Table 2 presents the quality of the virtual regions generated at different stages. For Qwen3, `Region valid` increases from 0.0207 at the Prompt stage to 0.9496 after SFT and reaches 0.9994 after GRPO. `Subject-cover@0.8` increases from 0.2162 to 0.8499, while `Cover+direction` increases from 0.3125 to 0.7970. This shows that SFT first enables the model to generate valid regions reliably, while complete training substantially improves subject coverage and directional consistency.

Table 2: Virtual region generation quality across different training stages.

Backbone	Metric	Prompt	SFT	GRPO
Qwen3-VL-8B-Instruct	Region valid	0.0207	0.9496	0.9994
	Subject-cover@0.5	0.4595	0.6739	0.9025
	Subject-cover@0.8	0.2162	0.5554	0.8499
	Cover+direction	0.3125	0.6229	0.7970
Qwen2.5-VL-7B-Instruct	Region valid	0.0241	0.7105	1.0000
	Subject-cover@0.5	0.6744	0.6950	0.9272
	Subject-cover@0.8	0.6512	0.5879	0.8751
	Cover+direction	0.5000	0.6205	0.8096
InternVL-8B	Region valid	0.0549	0.3253	0.9698
	Subject-cover@0.5	0.1939	0.7143	0.7038
	Cover+direction	0.1596	0.6560	0.6397

A similar trend is observed on Qwen2.5. Its `Region valid` increases from 0.0241 to 1.0000, `Subject-cover@0.5` from 0.6744 to 0.9272, and `Cover+direction` from 0.5000 to 0.8096. Notably, `Subject-cover@0.8` decreases slightly at the SFT stage but rises to 0.8751 after GRPO, indicating that GRPO is highly effective in guiding vision-language models to generate high-quality valid regions.

The region results on InternVL reveal a different pattern. Although `Region valid` reaches 0.9698 after GRPO, `Subject-cover` and `Cover+direction` do not consistently exceed their SFT-stage values. Together with the main results, this finding shows that generating a valid region does not necessarily mean that the model can effectively use it for subsequent localization. A gap remains between learning the format of intermediate states and achieving grounding gains.

4.6 Qualitative Analysis

Figure 4 presents three representative test samples. Each row shows the query text, ground truth, EAR, our SFT, and our GRPO predictions.

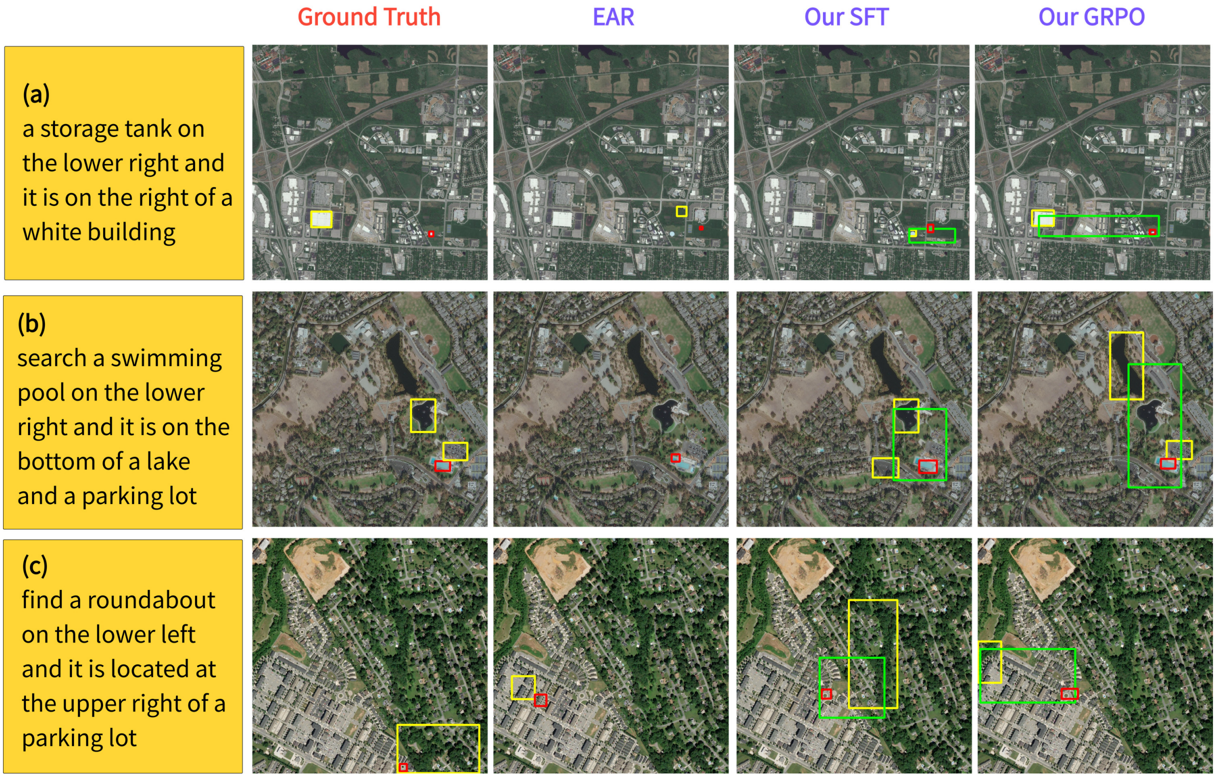

Figure 4: Qualitative comparison of EAR, VRT-SFT, and VRT-GRPO on representative test samples. Yellow boxes denote reference objects, green boxes denote virtual regions, and red boxes denote subjects.

In example (a), EAR localizes an area near the target, but its reference-object prediction does not accurately correspond to the white building in the ground truth. Within our VRT-GRPO framework, after SFT, the model can generate a virtual region covering the area around the target, but the reference-object localization remains inaccurate. After GRPO optimization, the model localizes the white building more accurately and generates a virtual region extending to its right and covering the storage tank, ultimately producing a subject prediction consistent with the ground truth. This example demonstrates how the object–region–subject trajectory makes the directional relationship between the reference entity and the target explicit.

In example (b), EAR fails to fully recover the multi-reference-object configuration in the query, whereas our SFT and our GRPO models explicitly generate multiple object boxes and the corresponding virtual regions. After GRPO optimization, the predicted region covers the target swimming pool and lies below the relevant reference entities, consistent with the relational constraints in the query. This example shows that the structured trajectory can represent the candidate search region jointly constrained by multiple reference objects and propagate this constraint to subsequent subject localization.

Example (c) reveals a limitation of the framework. Although both the SFT and GRPO models generate structurally valid virtual regions, their reference-object localization deviates from the ground-truth parking lot, causing the subsequent region and subject predictions to deviate further from the ground truth. This example shows that a valid region output alone does not guarantee correct grounding. When the reference entity is localized incorrectly, the error may propagate along the object–region–subject trajectory to the final target prediction.

4.7 Summary

Overall, the VRT-GRPO framework, which incorporates virtual regions and stage-specific process rewards, achieves higher multi-entity localization performance than the entity-aware baseline on Qwen3-VL and Qwen2.5-VL. The region-level results further show that, after training, the models can generate virtual regions with high validity, subject coverage, and directional consistency.

However, the performance degradation on InternVL shows that the benefits of the framework do not generalize uniformly across backbone models, and generating a valid spatial intermediate state does not necessarily improve final localization performance. Because the current comparison jointly changes the structured trajectory, virtual region, and process rewards, the results mainly support the effectiveness of the complete region-aware framework on selected backbone models, but they cannot yet determine the specific effect of the region intermediate state itself on the model's final localization.

5 Discussion

The overall experimental results show that VRT-GRPO improves multi-entity referring-expression grounding in remote sensing images on Qwen3-VL and Qwen2.5-VL, two models specifically tuned for instruction following [28, 29]. Compared with the entity-aware baseline, VRT-GRPO requires the model not only to predict the subject and reference objects but also to explicitly generate a virtual region, thereby transforming spatial relations in language into an intermediate search constraint in coordinate space. After SFT and GRPO training, the Qwen models can generate valid and directionally consistent virtual regions relatively reliably while achieving higher accuracy in final subject/object localization. This suggests that, for vision-language models with strong structured instruction-following capabilities, explicitly modeling intermediate spatial regions facilitates the step-by-step interpretation of complex spatial relations.

Meanwhile, the experiments also show that the benefits of VRT-GRPO are not consistent across all backbone models. Although InternVL-8B substantially increases the proportion of valid region outputs after GRPO, its final subject/object localization performance remains below that of the corresponding baseline. This indicates that generating a valid virtual region does not necessarily mean that the model can effectively use it for subsequent localization. In other words, a gap remains between learning the format of intermediate reasoning states and improving final grounding performance. Enhancing the ability of different backbone models to use the region for subject localization is therefore an important direction for future research.

Another limitation is that the current experiments validate the effectiveness of the complete VRT-GRPO framework rather than independently evaluating the contribution of the virtual region. Because VRT simultaneously introduces structured reasoning trajectories, virtual regions, and stage-specific process rewards, the current results cannot attribute the performance gains entirely to any single component. Future work could conduct more fine-grained ablation studies, such as removing the virtual region while keeping the plan and training budget unchanged, or separately disabling the region reward, direction reward, and plan reward, to more clearly analyze the contribution of each component.

Finally, the current GRPO rewards mainly focus on format, entity localization, region coverage, and directional consistency. Although they effectively improve the stability of structured outputs, they provide limited diagnostic capability for fine-grained errors in the reasoning process. Future work could introduce more detailed process supervision or automated judging mechanisms [18] to model error types such as incorrect reference-object selection, incorrect region direction, and inconsistency between the subject and region separately, thereby improving the specificity and interpretability of the training signals.

Overall, VRT-GRPO demonstrates the potential of introducing explicit intermediate spatial regions into remote sensing localization under complex spatial constraints. If future work can further improve the ability of different backbone models to use virtual regions and refine the method through more fine-grained ablations and more flexible region representations, the framework may achieve more stable performance on more complex remote sensing spatial-reasoning tasks.

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A Implementation Details and Prompt Templates

This appendix presents the main implementation details used in our experiments, including the prompt templates for Virtual Region Trajectory, the construction rules for virtual-region pseudo-annotations, the SFT data construction procedure, the GRPO reward functions, training hyper-parameters, and inference settings.

A.1 VRT Reasoning Prompt

The following VRT structured-output template is used at the Prompt, SFT, and GRPO stages. The model is required to generate the `<plan>`, `<think>`, and `<answer>` sections in sequence and to explicitly annotate the reference objects, virtual region, and subject within the reasoning trajectory.

```
Localize objects in the image based on the following description: {instruction}

[Global Structure Framework]
You must output your response using exactly this structure:
1. <plan>: Concisely generate a step-by-step reasoning plan based on the description.
2. <think>: Follow the plan to detail your search process. Whenever an object, region, or
    subject is identified, tag it naturally in your sentences like <object> [x1, y1, x2, y2] or
    <region> [x1, y1, x2, y2] or <subject> [x1, y1, x2, y2].
3. <answer>: Summarize the box coordinates from the think block.

[Hard Rules]
1. Concept Distinction: Use "object" for existing reference entities, "region" for deduced
    virtual spaces, and "subject" for the target.
2. Answer Consistency: The box coordinates within the final <answer> must be exactly identical
    to those in the <think> block.
3. You MUST ALWAYS have at least one bounding box for <subject>.
```

Here, {`instruction`} denotes the original language query after removing the `<image>` tag. All coordinates use the pixel coordinate system of the original image.

A.2 SFT Teacher Prompt

When constructing the SFT data, InternVL is first used to generate structured reasoning trajectories [30]. To prevent the teacher model from directly generating unstable coordinates, it is required to output placeholders only. The program then replaces `OBJECT_i`, `REGION`, and `SUBJECT` with the corresponding ground-truth or pseudo-annotated boxes.

```
Localize objects in the image based on the following description: {instruction}

[Global Structure Framework]
You must output your response using exactly this structure:
1. <plan>: Concisely generate a step-by-step reasoning plan based on the description.
2. <think>: Follow the plan to detail your search process. Whenever an object, region, or
    subject is identified, tag it naturally in your sentences like <object> OBJECT_1 or <region>
     REGION or <subject> SUBJECT.
3. <answer>: Summarize the placeholders from the think block.

[Allowed Placeholders]
Objects: OBJECT_1, ..., OBJECT_N
Region: REGION
Subject: SUBJECT

[Hard Rules]
1. Concept Distinction: Use "object" for existing reference entities, "region" for deduced
    virtual spaces, and "subject" for the target.
```

```
2. Answer Consistency: The placeholders within the final <answer> must be exactly identical to
   those in the <think> block.
3. You MUST ALWAYS have one placeholder for <subject>.
4. Do NOT output numeric coordinates. Use placeholders only.
5. Do NOT invent extra placeholders.
6. Do NOT output JSON, Markdown, or extra explanations outside the three required parts.
```

After the teacher output passes the format checks, the program replaces the placeholders with the corresponding gt_anchor_boxes, pseudo_region_box, and gt_subject_box to obtain the final SFT training samples.

A.3 External Critic Prompt

The planning reward in GRPO is provided by an external VLM/text critic. The critic does not receive the image. It only evaluates whether <plan> covers the query constraints and whether <think>/<answer> follows the plan.

```
You are a strict and objective text-logic judge.

Your task is to evaluate whether the model-generated `<plan>` covers the constraints in the user
    instruction, and whether the actual execution trajectory follows that plan.

You will not see the image. Do not evaluate coordinate accuracy, bbox correctness, or output
   format.
Only evaluate the following two text-level dimensions:

1. constraint_coverage
Evaluate whether the Plan covers the target, reference objects, attributes, and spatial
   relations in the user instruction.
- 9 points: Fully covers the key constraints.
- 6 points: Covers the main constraints, but misses a few minor conditions.
- 3 points: Misses key constraints and may lead to the wrong region.
- 0 points: Barely covers the instruction, or identifies the wrong target category.

2. execution_consistency
Evaluate whether the execution trajectory involving object, region, and target generally follows
    the Plan.
- 9 points: The execution is highly consistent with the Plan.
- 6 points: Mostly consistent, but with minor omissions or unclear correspondence.
- 3 points: The execution is clearly disconnected from the Plan.
- 0 points: The execution largely ignores the Plan.

Strictly output JSON only. Do not use Markdown or provide any extra explanation:

output json:
{
  "reasoning": "Briefly explain the reason for any deductions",
  "constraint_coverage": <integer>,
  "execution_consistency": <integer>
}
```

The planning reward is defined as

$$R_{\text{plan}} = \frac{s_c + s_e - 12}{6}, \tag{A.1}$$

where s_c and s_e denote constraint_coverage and execution_consistency, respectively, and both range from 0 to 9. If <plan> is missing, an unweighted reward of -1.0 is assigned directly.

B Virtual-Region Pseudo-Annotation Construction

Given the ground-truth subject box b_s^* and the set of ground-truth reference-object boxes

$$\mathcal{B}_o^* = \{b_o^{i*}\}_{i=1}^N, \tag{B.1}$$

we construct the virtual-region pseudo-annotation b_r^* for an image of width and height (W, H) using deterministic geometric rules.

We first compute the union box of the reference objects:

$$b_o^u = \mathrm{BBox}(\mathcal{B}_o^*), \tag{B.2}$$

and the union box of the subject and reference objects:

$$b_u = \mathrm{BBox}(\{b_s^*\} \cup \mathcal{B}_o^*). \tag{B.3}$$

We then compute the center point of the subject and that of the reference-object union box:

$$c_s = \mathrm{Center}(b_s^*), \qquad c_o = \mathrm{Center}(b_o^u). \tag{B.4}$$

If the subject and the reference-object union box do not overlap along the x-axis, horizontal extension is enabled. If they do not overlap along the y-axis, vertical extension is enabled. If they overlap along both axes, the axis with the larger center offset is selected as the primary direction:

$$d = \begin{cases} x, & |c_s^x - c_o^x| \geq |c_s^y - c_o^y|, \\ y, & \text{otherwise.} \end{cases} \tag{B.5}$$

Along each enabled direction, the region extends from the union box toward the side on which the subject lies relative to the reference object, up to the image boundary. For example, when $c_s^x \geq c_o^x$, the subject lies to the right of the reference object, so the right boundary of the region is set to W; otherwise, the left boundary is set to 0. The same rule applies in the vertical direction.

Finally, padding and clipping are applied to the resulting region. The default padding ratio is $\lambda = 0.15$, while the padding ratio on the side opposite the primary direction is $\lambda_{\mathrm{opp}} = 0.0$, meaning that no additional extension is applied behind the anchor. The final result is clipped to the image boundaries:

$$b_r^* = \mathrm{Clip}\Big(\mathrm{Pad}(\tilde{b}_r, \lambda, \lambda_{\mathrm{opp}}), W, H\Big). \tag{B.6}$$

If a sample lacks a valid reference object, the procedure falls back to applying symmetric padding to the subject box.

C SFT Data Construction

SFT data construction consists of three steps. First, a `pseudo_region_box` is constructed from the ground-truth subject and reference-object annotations. Second, InternVL is used to generate a VRT teacher trajectory containing placeholders. Third, the placeholders in the teacher trajectory are replaced with the ground-truth coordinates.

Each SFT sample is stored in chat format:

```
{
  "messages": [
    {
      "role": "user",
      "content": [
        {"type": "image"},
        {"type": "text", "text": "VRT prompt"}
      ]
    },
    {
      "role": "assistant",
      "content": "<plan>...</plan><think>...</think><answer>...</answer>"
    }
  ],
  "gt_subject_box": "...",
  "gt_anchor_boxes": "...",
  "sft_region_box": "..."
}
```

Invalid samples are filtered during construction. A valid sample must satisfy the following conditions: it contains a valid subject box; it contains at least one valid reference-object box; it contains a valid `pseudo_region_box`; the teacher output contains `<plan>`, `<think>`, and `<answer>`; the numbering of `OBJECT_i` in the teacher output matches the actual number of objects; and both the `REGION` and `SUBJECT` placeholders are present.

Greedy decoding is used during teacher generation:

$$\texttt{do\_sample=False}, \qquad \texttt{max\_new\_tokens} = 512. \tag{C.1}$$

D Reward Function Implementation Details

Five types of reward terms are used during the GRPO stage. The total reward is

$$R = 0.15R_{\text{plan}} + 0.25R_{\text{fmt}} + 1.00R_{\text{ent}} + 0.25R_{\text{reg}} + 0.20R_{\text{spa}}. \tag{D.1}$$

Here, R_{plan} is the planning reward, R_{fmt} is the format and logic reward, R_{ent} is the entity-localization reward, R_{reg} is the virtual-region reward, and R_{spa} is the spatial-consistency reward.

D.1 Format and Logic Reward

The format reward consists of structural completeness, count constraints, and coordinate consistency:

$$R_{\text{fmt}} = R_{\text{str}} + R_{\text{num}} + R_{\text{cons}}. \tag{D.2}$$

If the output strictly follows the order `<plan>`, `<think>`, and `<answer>`, then

$$R_{\text{str}} = 1.0. \tag{D.3}$$

Otherwise, the presence of each of the three tags is checked separately:

$$R_{\text{str}} = \sum_{t \in \{\text{plan},\text{think},\text{answer}\}} r_t, \qquad r_t = \begin{cases} 0.15, & t \text{ exists}, \\ -0.25, & t \text{ does not exist}. \end{cases} \tag{D.4}$$

Let the numbers of subjects, regions, and objects parsed from <think> be n_s, n_r, and n_o, respectively, and let the number of ground-truth reference objects be N. Then,

$$R_{\text{num}} = \phi_s(n_s) + \phi_r(n_r) + \phi_o(n_o, N), \tag{D.5}$$

where

$$\phi_s(n_s) = \begin{cases} 0.50, & n_s = 1, \\ -0.75, & n_s \neq 1, \end{cases} \qquad \phi_r(n_r) = \begin{cases} 0.40, & n_r = 1, \\ -0.60, & n_r \neq 1, \end{cases} \tag{D.6}$$

and

$$\phi_o(n_o, N) = \begin{cases} 0.40, & n_o = N, \\ -0.40 - 0.10|n_o - N|, & n_o \neq N. \end{cases} \tag{D.7}$$

The consistency reward checks whether the subject, object, and region coordinates in <think> and <answer> are consistent:

$$R_{\text{cons}} = \sum_{z \in \{\text{sub,obj,reg}\}} \psi_z, \qquad \psi_z = \begin{cases} 0.20, & \hat{B}_z^{\text{think}} = \hat{B}_z^{\text{answer}}, \\ -0.40, & \text{otherwise.} \end{cases} \tag{D.8}$$

D.2 Entity-Localization Reward

The entity reward includes subject localization and reference-object localization:

$$R_{\text{ent}} = R_{\text{sub}} + R_{\text{obj}}. \tag{D.9}$$

For the subject, if the predicted box is valid, let

$$u_s = \text{IoU}(\hat{b}_s, b_s^*), \tag{D.10}$$

and compute

$$R_{\text{sub}} = 2.0u_s + \eta_s(u_s), \tag{D.11}$$

where

$$\eta_s(u_s) = \begin{cases} 0.50, & u_s \geq 0.7, \\ 0.30, & 0.5 \leq u_s < 0.7, \\ -0.40, & u_s < 0.1, \\ 0, & \text{otherwise.} \end{cases} \tag{D.12}$$

If the subject box is invalid, a penalty of -1.0 is assigned. If exactly one subject is parsed, an additional reward of 0.20 is assigned. If the subject coordinates in <think> and <answer> are consistent, another reward of 0.20 is assigned; otherwise, a penalty is applied.

For reference objects, greedy one-to-one IoU matching is first performed between the predicted and ground-truth object boxes. Let the matched IoUs be $\{u_o^i\}_{i=1}^N$. Then,

$$R_{\text{obj}} = \frac{1}{N} \sum_{i=1}^{N} \left(1.2u_o^i + \eta_o(u_o^i)\right), \tag{D.13}$$

where

$$\eta_o(u_o) = \begin{cases} 0.30, & u_o \geq 0.7, \\ 0.20, & 0.5 \leq u_o < 0.7, \\ -0.20, & u_o < 0.1, \\ 0, & \text{otherwise.} \end{cases} \tag{D.14}$$

If the number of predicted objects equals the ground-truth count, an additional reward of 0.20 is assigned; otherwise, a penalty of $-0.40|n_o - N|$ is applied according to the count difference.

D.3 Virtual-Region Reward

If the number of parsed regions is not one, then

$$R_{\text{reg}} = -1.5 - 0.25|n_r - 1|. \tag{D.15}$$

If exactly one valid region exists, let

$$u_r = \mathrm{IoU}(\hat{b}_r, b_r^*), \qquad c_r = \mathrm{IoC}(b_s^*, \hat{b}_r), \tag{D.16}$$

where IoC denotes the proportion of the subject covered by the region. Then,

$$R_{\text{reg}} = 1.5u_r + 0.5c_r + \eta_r(u_r, c_r), \tag{D.17}$$

with

$$\eta_r(u_r, c_r) = \eta_{\text{iou}}(u_r) + \eta_{\text{cov}}(c_r), \tag{D.18}$$

$$\eta_{\text{iou}}(u_r) = \begin{cases} 0.50, & u_r \geq 0.5, \\ -0.50, & u_r < 0.1, \\ 0, & \text{otherwise}, \end{cases} \qquad \eta_{\text{cov}}(c_r) = \begin{cases} -0.50, & c_r < 0.5, \\ 0, & c_r \geq 0.5. \end{cases} \tag{D.19}$$

D.4 Spatial-Consistency Reward

If the number of parsed regions is not one, then

$$R_{\text{spa}} = -1.25. \tag{D.20}$$

Otherwise, the subject coverage is computed as

$$c_r = \mathrm{IoC}(b_s^*, \hat{b}_r). \tag{D.21}$$

The coverage reward is

$$R_{\text{cov}} = c_r + \eta_{\text{spa}}(c_r), \qquad \eta_{\text{spa}}(c_r) = \begin{cases} 0.50, & c_r \geq 0.8, \\ -0.75, & c_r < 0.5, \\ 0, & \text{otherwise}. \end{cases} \tag{D.22}$$

The set of directional dimensions $\mathcal{D}(q)$ is then determined from the directional terms in the query. In the training reward, the horizontal dimension is included when the query contains *left* or *right*, while the vertical dimension is included when the query contains *above*, *below*, *upper*, *lower*, *top*, *bottom*, *over*, or *under*.

For each reference object, we determine whether the predicted region and the ground-truth subject lie in the same relational direction from that object:

$$a_i = \frac{1}{|\mathcal{D}(q)|} \sum_{d \in \mathcal{D}(q)} \mathbb{I}\Big[\big(c_d(\hat{b}_r) - c_d(b_o^{i*})\big)\big(c_d(b_s^*) - c_d(b_o^{i*})\big) > 0\Big]. \tag{D.23}$$

If valid directional comparisons exist, then

$$R_{\text{spa}} = R_{\text{cov}} + 0.5\left(2 \cdot \frac{1}{N}\sum_{i=1}^{N} a_i - 1\right). \tag{D.24}$$

If the query contains no parseable directional dimension, only the coverage reward is used.

D.5 GRPO Advantage Function

For each input x_i, GRPO samples $K = 8$ candidate outputs $\{y_{i,k}\}_{k=1}^{K}$ and computes their corresponding rewards $R_{i,k}$ [16]. We use within-group centering and batch-level variance scaling:

$$A_{i,k} = \frac{R_{i,k} - \frac{1}{K}\sum_{j=1}^{K} R_{i,j}}{\sigma_{\mathcal{B}} + \epsilon}, \tag{D.25}$$

where $\sigma_{\mathcal{B}}$ denotes the standard deviation of the rewards of all candidate outputs in the current batch, and ϵ is a numerical-stability term.

The policy-optimization objective uses a clipped policy objective with a KL constraint:

$$\begin{aligned}\mathcal{L}_{\mathrm{GRPO}}(\theta) = -\mathbb{E}_{i,k}\Big[&\min\big(\rho_{i,k}(\theta)A_{i,k}, \mathrm{clip}(\rho_{i,k}(\theta), 1-\varepsilon, 1+\varepsilon)A_{i,k}\big)\\ &- \beta D_{\mathrm{KL}}(\pi_\theta \,\|\, \pi_{\mathrm{ref}})\Big],\end{aligned} \tag{D.26}$$

where $\rho_{i,k}(\theta)$ is the policy probability ratio, $\beta = 0.0025$, and the reference policy is the LoRA adapter frozen at SFT initialization.

E Training and Inference Settings

E.1 SFT Settings

Table E.1: SFT training settings.

Parameter	Value
LoRA rank	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`
Learning rate	1×10^{-4}
Max steps	200
Per-device batch size	1
Gradient accumulation	16; automatically scaled by the number of GPUs under `torchrun`
Max model length	2048
Warmup ratio	0.05
Optimizer	AdamW
Precision	`bfloat16`
Save steps	50

E.2 GRPO Settings

Table E.2: GRPO training settings.

Parameter	Value
Number of generations (K)	8
Learning rate	1×10^{-5}
Epochs	1
Per-device batch size	1
Expected GPUs	4
Target generation batch size	32
Max completion length	512
Generation temperature	0.8
Top-p	0.95
Warmup ratio	0.01
KL beta	0.0025
Max grad norm	1.0
Reward scaling	`batch`
Loss type	DAPO [32]
Mask truncated completions	`True`
Disable dropout	`True`
Optimizer	AdamW fused
Seed	42

GRPO training samples must contain valid subject, reference-object, and pseudo-region boxes, and the pseudo-region must almost completely cover the subject:

$$\mathrm{IoC}(b_s^*, b_r^*) \geq 0.999. \tag{E.1}$$

E.3 External Critic Settings

Table E.3: External critic settings.

Parameter	Value
Critic model	`qwen3vl-30b-critic`
API format	OpenAI-compatible chat completions
Temperature	0
Max tokens	256
Timeout	60 s
Retries	2

The critic receives only the instruction, plan, and execution trajectory in textual form. It does not receive the image or evaluate coordinate accuracy.

E.4 Inference Settings

VRT inference at the Prompt, SFT, and GRPO stages uses greedy decoding:

$$\texttt{do\_sample=False}. \tag{E.2}$$

The maximum generation length for VRT inference is

$$\texttt{max\_new\_tokens} = 2048. \tag{E.3}$$

The maximum generation length for entity-aware baseline inference is

$$\texttt{max\_new\_tokens} = 1024. \tag{E.4}$$

All output coordinates are parsed and evaluated in the pixel coordinate system of the original image. If the model outputs multiple boxes of the same type, the evaluation script first attempts to parse them from the <think> tag. If parsing from <think> fails, it parses them from <answer>.

F Region Evaluation Metrics

In addition to the final subject/object grounding metrics, we evaluate the generation quality of the virtual region. We first measure whether the model generates a valid region:

$$\mathrm{RegionValid} = \frac{1}{M}\sum_{i=1}^{M} \mathbb{I}[\hat{b}_{r,i} \text{ is valid}]. \tag{F.1}$$

For samples from which a valid region can be parsed, subject coverage is computed as

$$\text{Subject-cover@}\tau = \frac{\sum_{i=1}^{M} \mathbb{I}\left[\mathrm{IoC}(b^*_{s,i}, \hat{b}_{r,i}) \geq \tau \wedge \hat{b}_{r,i} \text{ is valid}\right]}{\sum_{i=1}^{M} \mathbb{I}[\hat{b}_{r,i} \text{ is valid}]}. \tag{F.2}$$

We report results at $\tau = 0.5$ and $\tau = 0.8$.

The directional-consistency metric determines whether the predicted region and the ground-truth subject lie in the same direction relative to the reference object. Let the directional-consistency score be a_i. Then,

$$\mathrm{DirectionConsistent} = \frac{1}{M_d}\sum_{i} \mathbb{I}[a_i \geq 1.0], \tag{F.3}$$

where M_d denotes the number of samples with a parseable directional relation. In the evaluation script, the directional terms include *left*, *right*, *east*, *west*, *above*, *below*, *upper*, *lower*, *top*, *bottom*, *over*, *under*, *north*, and *south*.

The joint metric `Cover+Direction` requires the region to both cover the subject and satisfy directional consistency:

$$\text{Cover+Direction} = \frac{1}{M_c}\sum_{i} \mathbb{I}\left[\mathrm{IoC}(b^*_{s,i}, \hat{b}_{r,i}) \geq 0.5 \wedge a_i \geq 1.0\right]. \tag{F.4}$$

This metric measures whether the virtual region simultaneously satisfies the two conditions of “containing the target search area” and “conforming to the relative spatial direction.”